

HerPlate Ghana: Policy & Implementation

A Stakeholder Operational Guide to De-Bottlenecking Women's Nutrition in Ghana.

- **Initiative:** Women in Data™ 2026 Datathon — EAT Track
- **Team:** Plateful of Data
- **Live Tool:** herplate-ghana.streamlit.app
- **Repository:** [GitHub Repository](#)

Summary & Core Principle

Traditional food security strategies treat low dietary intake as a uniform issue, addressing it through generic farm production subsidies or nutrition awareness campaigns. However, **HerPlate Ghana** demonstrates that food systems fail through fundamentally different mechanisms: **"Same dietary gap. Different bottlenecks."**

While Ghana produces over 3,100 kcal per capita daily, **65.8% of the population (23.1 million people)** cannot afford a standard healthy diet. Concurrently, **50.1% of reproductive-age women** fail to attain Minimum Dietary Diversity (MDD-W). This playbook translates HerPlate's multi-signal diagnostics into actionable operational protocols for national policymakers, public health authorities, and logistics coordinators.

1. Diagnostic Architecture: The Tri-Signal Matrix

Food Group	Intake (2022)	Supply Shift (2012–2022)	Cost Share (2021)	Bottleneck Archetype	System Failure Mechanism
Dairy	15.4% (8.1% Rural / 20.9% Urban)	-24.4% (8.2 → 6.2 kg/cap/yr)	41.1% (\$1.44 PPP/day ASF)	Convergent Pressure	Dual Contraction & Cost Barrier: Severe intake deficit driven simultaneously by shrinking per-capita supply and prohibitive retail prices.
Pulses	23.5% (25.5% Rural / 22.0% Urban)	-39.4% (16.5 → 10.0 kg/cap/yr)	11.1% (\$0.39 PPP/day)	Supply/Diversity Pressure	Upstream Harvest Collapse: Physical scarcity from domestic yield declines drives low intake; retail prices

					remain relatively accessible.
Eggs	27.6% (18.3% Rural / 34.7% Urban)	+3.9% (1.81 → 1.88 kg/cap/yr)	41.1% (Parent ASF Basket)	Access Beyond Supply	Post-Farmgate Logistics Chokepoint: Stable aggregate production fails to reach rural plates due to cold-chain absence and transport spoilage.
Vegetables	71.8% (Leafy) / 51.5% (Other)	-40.7% (54.8 → 32.5 kg/cap/yr)	13.1% (\$0.46 PPP/day)	Emerging Supply Concern	Upstream Buffer Depletion: Strong traditional soup consumption sustains current intake, but a ~41% harvest contraction creates acute risk.

2. Operational Action Protocols by Archetype

Protocol A: De-bottlenecking Perishables & Eggs (Access Beyond Supply)

- **Lead Stakeholders:** Supply-Chain Planners, Market Queens, Ghana National Egg Producers Association (GNEPA).
- **Failure Point:** Production is stable (+3.9%), but a **16.4 percentage point urban-rural access divide** exists due to last-mile transport losses and ambient temperature spoilage.
- **Action Steps:**
 1. *Halt Upstream Flock Expansion Subsidies:* Divert public capital away from generic hatchery expansion.
 2. *Deploy Village-Level Solar Refrigeration:* Finance modular, off-grid solar cold hubs at rural aggregation centers.
 3. *Establish Farmgate-to-Market Shuttles:* Partner with regional transport unions to schedule consolidated, ventilated morning deliveries to rural retail kiosks.

Protocol B: Restoring Domestic Legumes & Pulses (Supply/Diversity Pressure)

- **Lead Stakeholders:** Ministry of Food and Agriculture (MoFA), Agricultural Extension Agents (AEAs).
- **Failure Point:** Cowpea and bean availability fell by **-39.4%**, restricting diet diversity despite remaining economically affordable (11.1% basket share).

- **Action Steps:**
 1. **Certified Seed Subsidies:** Distribute drought-resilient, early-maturing cowpea and pigeon pea seed packages directly to smallholders.
 2. **Inoculant & Soil-Health Kits:** Provide rhizobium inoculants to increase nitrogen fixation and lower cultivation overhead.
 3. **Targeted Farmgate Procurement:** MoFA and local school feeding coordinators establish forward procurement contracts at guaranteed floor prices.

Protocol C: Mitigating High-Cost Animal Protein & Dairy (Convergent Pressure)

- **Lead Stakeholders:** National Development Planning Commission (NDPC), Ghana Health Service (GHS), Ministry of Trade.
- **Failure Point:** Animal-source foods comprise **41.1% of diet costs**, while milk availability fell by **24.4%**, resulting in an 8.1% rural consumption rate.
- **Action Steps:**
 1. *Deploy Targeted Consumer Price Vouchers:* Issue targeted digital vouchers for vulnerable pregnant/lactating women in rural primary health clinics.
 2. *Decentralized Milk Chilling Nodes:* Co-locate livestock milk collection hubs near pastoral transit corridors to reduce transit spoilage.
 3. *Tariff Stabilization on Core Dairy Inputs:* Adjust import duties on essential dairy feed formulations to prevent feed cost spikes.

Protocol D: Protecting Fresh Produce Buffers (Emerging Supply Concern)

- **Lead Stakeholders:** MoFA Horticulture Division, Irrigation Development Authority.
- **Failure Point:** Per-capita vegetable harvests collapsed by **-40.7%**, threatening traditional leafy green dietary buffers.
- **Action Steps:**
 1. *Dry-Season Drip Irrigation Grants:* Install smallholder micro-irrigation systems in peri-urban and riverine farming belts.
 2. *Post-Harvest Produce Crating:* Replace traditional overpacked sacks with standardized plastic crates during transport to reduce physical crop loss.

3. Cross-Sector Governance & Coordination Model

To ensure policy sustainability, interventions must be coordinated across three institutional levels:

- 1) **POLICY LEVEL:** MoFA & National Planning Commission (NDPC)
 - Reallocate subsidies from calorie grains to pulse seeds.
 - Anchor national dietary targets into budgetary frameworks.
- 2) **CLINICAL LEVEL:** Ghana Health Service & Maternal Clinics
 - Integrate HerPlate disparity indicators into MDD-W tracking.

- Target clinical food assistance based on local bottlenecks.
- 3) LOGISTICS LEVEL: Cold-Chain Operators & Producer Unions
- Manage decentralized rural solar storage hubs.
 - Eliminate last-mile distribution margins for perishables.

4. Technical Replication & Scenario Modeling

Program officers and evaluators can replicate this analysis and simulate policy interventions via the open codebase:

Clone and Install:

```
git clone https://github.com/Nowshin1077/Women_in_Data_Datathon_Project_2026.git  
cd Women_in_Data_Datathon_Project_2026  
pip install -r requirement.txt
```

Execute data pipeline:

```
python src/final_edu.py
```

Launch the interface:

```
streamlit run src/app.py
```